

Year 7 · Chapter 7 · Decimals: answers (teacher copy)

7.01 Ordering decimals (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: 0.7
- Ex 1b: 34.89
- Ex 2a: 8 hundredths
- Ex 2b: 3.998, 34.89, 34.98, 36.86, 38.141, 340
- Ex 3a: 1.293, 1.239, 1.1015, 1.015, 1.006
- Ex 3b: e.g. 2.355

Easy · Set A (quick drill)

- 0.7
- 2.5
- 0.1
- 14.3
- 5.6
- 0.34
- 0.8
- 1.2
- 0.1
- 3.5
- 7.7
- 0.55

Easy · Set B

- 7 tenths
- 7 hundredths
- 7 thousandths
- 7 tens
- 7 thousandths
- 7 ones

Medium · Set C

- 3.998, 34.89, 34.98, 36.86, 38.141, 340
- 0.045, 0.405, 0.45, 0.5, 0.54
- 2.01, 2.011, 2.1, 2.101, 2.11
- 7.06, 7.066, 7.6, 7.606, 7.66

Challenging · Set D

- 1.293, 1.239, 1.1015, 1.015, 1.006
- e.g. 2.355
- 12.018 s
- 0.1: 0.100 000 > 0.099 999

Extension

- 0.605
- 9 (4.31 to 4.39)

Exit ticket

- Level 1: 0.4
- Level 2: 5.02, 5.2, 5.22
- Level 3: 0.33, 0.303, 0.3, 0.033

7.02 Decimals and fractions (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: $\frac{1}{2}$
- Ex 1b: $\frac{89}{100}$
- Ex 2a: $\frac{111}{250}$
- Ex 2b: 0.4
- Ex 3a: 0.07
- Ex 3b: 0.011

Easy · Set A (quick drill)

- $\frac{3}{10}$
- $\frac{89}{100}$
- $\frac{9}{100}$
- $\frac{7}{10}$
- $\frac{21}{100}$
- $\frac{3}{1000}$
- $\frac{2}{5}$
- $\frac{1}{4}$
- $\frac{3}{4}$
- $\frac{3}{50}$
- $\frac{1}{8}$
- $\frac{7}{20}$

Easy · Set B

- 0.4
- 0.13
- 0.07
- 0.011
- 0.9
- 0.125

Medium · Set C

- $\frac{1}{2}$
- $\frac{1}{4}$
- $\frac{111}{250}$
- $\frac{2}{25}$
- $2\frac{3}{5}$
- $3\frac{1}{8}$

Challenging · Set D

- 0.6
- 0.35
- 0.036
- $\frac{3}{8} = 0.375$ is larger

Extension

- 0.075
- $4\frac{3}{40}$

Exit ticket

- Level 1: $\frac{7}{10}$
- Level 2: 0.23
- Level 3: $\frac{7}{20}$

7.03 Adding and subtracting decimals

(booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: 16.88
2. Ex 1b: 2.379
3. Ex 2a: 27.595
4. Ex 2b: 104.49
5. Ex 3a: 396.51
6. Ex 3b: 65.355

Easy · Set A (quick drill)

1. 5.9
2. 16.88
3. 1.5
4. 4.5
5. 7.3
6. 5.75
7. 4.7
8. 3.5
9. 1.3
10. 3.6
11. 3.95
12. 6.75
13. 6.25
14. 0.98

Easy · Set B

1. 2.379
2. 5.18
3. 4.55
4. 12.03
5. 23.568
6. 19.98

Medium · Set C

1. 104.49
2. 1 037.9
3. 11.315
4. 177.595
5. 252.62
6. 5.544

Challenging · Set D

1. 65.355
2. 396.51
3. \$29.75
4. 0.85 kg

Extension

1. 8.43
2. $6.42 + 5.31 = 11.73$ (or similar)

Exit ticket

1. Level 1: 4.3
2. Level 2: 6.55
3. Level 3: 19.08

7.04 Multiplying and dividing by powers of 10

(booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: 75.4
2. Ex 1b: 754
3. Ex 2a: 7 540
4. Ex 2b: 1.39
5. Ex 3a: 0.139
6. Ex 3b: 0.0139

Easy · Set A (quick drill)

1. 75.4
2. 754
3. 3
4. 4 820
5. 5.6
6. 125
7. 36
8. 45
9. 2 100
10. 4.8
11. 0.057
12. 0.36
13. 9
14. 0.12

Easy · Set B

1. 1.39
2. 0.139
3. 0.6
4. 0.0452
5. 0.008
6. 0.35

Medium · Set C

1. 0.0139
2. 7.2
3. 25 000
4. 0.00091
5. 4
6. 850

Challenging · Set D

1. \$135
2. \$8.75
3. 0.096 mm
4. 1000

Extension

1. 7 200
2. 0.16

Exit ticket

1. Level 1: 38
2. Level 2: 0.526
3. Level 3: 49

7.05 Using a known product

(booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: 319.2
2. Ex 1b: 31.92
3. Ex 2a: 319.2
4. Ex 2b: 3.192
5. Ex 3a: 319.2
6. Ex 3b: 0.3192

Easy · Set A (quick drill)

1. 319.2
2. 319.2
3. 31.92
4. 31 920
5. 31.92
6. 31.92
7. 32.2
8. 32.2
9. 3.22
10. 3 220
11. 3.22
12. 0.322

Easy · Set B

1. 34.5
2. 3.45
3. 3.45
4. 3.45
5. 345
6. 0.0345

Medium · Set C

1. 319.2
2. 3.192
3. 0.3192
4. 3.192
5. 31.92
6. 42

Challenging · Set D

1. 45
2. 45
3. 0.045
4. 12.5

Extension

1. 17
2. 24

Exit ticket

1. Level 1: 32.2
2. Level 2: 3.22
3. Level 3: 0.0322

7.06 Multiplying decimals (booklet p. 23)**WE DO examples (teacher)**

1. Ex 1a: 16.5
2. Ex 1b: 0.6
3. Ex 2a: 7.32
4. Ex 2b: 1.3325
5. Ex 3a: 0.001
6. Ex 3b: 5.137

Easy · Set A (quick drill)

1. 16.5
2. 4.6
3. 29.1
4. 288.92
5. 1.8
6. 12
7. 1.2
8. 3.6
9. 0.08
10. 10
11. 0.3
12. 7.7
13. 0.21
14. 0.42

Easy · Set B

1. 0.6
2. 0.12
3. 0.63
4. 0.5
5. 0.001
6. 1.21

Medium · Set C

1. 7.32
2. 1.3325
3. 5.137
4. 4.34
5. 2.1
6. 0.543

Challenging · Set D

1. 14.4 m²
2. \$80.33
3. 0.008
4. less: multiplying by less than 1 makes it smaller

Extension

1. 0.0081
2. e.g. 0.4×0.6

Exit ticket

1. Level 1: 8.6
2. Level 2: 0.42
3. Level 3: 3.29

7.07 Dividing decimals by whole numbers

(booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 381.2
2. Ex 1b: 12.2
3. Ex 2a: 65.2
4. Ex 2b: 0.6875
5. Ex 3a: 0.4375
6. Ex 3b: \$8.40

Easy · Set A (quick drill)

1. 381.2
2. 2.1
3. 3.1
4. 0.25
5. 0.09
6. 2.1
7. 2.4
8. 2.3
9. 2.1
10. 0.07
11. 1.2
12. 0.3
13. 0.8
14. 0.2

Easy · Set B

1. 12.2
2. 65.2
3. 0.91
4. 0.08
5. 7.8
6. 0.9

Medium · Set C

1. 0.6875
2. 0.4375
3. 1.75
4. 0.15
5. 2.42
6. 0.15

Challenging · Set D

1. \$8.40
2. 1.125 m
3. \$1.45
4. 4.22 km

Extension

1. 5.6
2. 1.25 and 0.625: doubling the divisor halves the answer

Exit ticket

1. Level 1: 3.4
2. Level 2: 2.4
3. Level 3: 1.325

7.08 Dividing by a decimal (booklet p. 31)**WE DO examples (teacher)**

1. Ex 1a: 25
2. Ex 1b: 31.8
3. Ex 2a: 230
4. Ex 2b: 25.4
5. Ex 3a: 5
6. Ex 3b: 14 cups

Easy · Set A (quick drill)

1. 25
2. 30
3. 9
4. 8
5. 6
6. 20
7. 6
8. 12
9. 5
10. 2
11. 70
12. 5
13. 30
14. 30

Easy · Set B

1. 31.8
2. 4.9
3. 32
4. 0.8
5. 30
6. 10

Medium · Set C

1. 230
2. 25.4
3. 5
4. 49
5. 600
6. 0.42

Challenging · Set D

1. 14 cups
2. 12.5 km/L
3. 14 m
4. there are 10 halves in 5

Extension

1. 180
2. the same: both are 80

Exit ticket

1. Level 1: 20
2. Level 2: 9
3. Level 3: 4

7.09 Fractions to decimals (booklet p. 35)**WE DO examples (teacher)**

1. Ex 1a: 0.8
2. Ex 1b: 0.375
3. Ex 2a: $0.\overset{\cdot}{6}$
4. Ex 2b: $0.\overset{\cdot}{5}$
5. Ex 3a: $0.\overset{\cdot}{4}\overset{\cdot}{5}$
6. Ex 3b: $0.\overset{\cdot}{1}\overset{\cdot}{6}$

Easy · Set A (quick drill)

1. 0.8
2. 0.25
3. 0.75
4. 0.5
5. 0.4
6. 0.35
7. 0.5
8. 0.6
9. 0.25
10. 0.7
11. 0.125
12. $0.\dot{3}$
13. 0.45
14. $0.\dot{2}$

Easy · Set B

1. 0.375
2. 0.625
3. 0.0625
4. 0.225
5. 0.875
6. 0.12

Medium · Set C

1. $0.\dot{6}$
2. $0.\dot{5}$
3. $0.\dot{3}$
4. $0.\dot{4}5$
5. $0.\dot{1}6$
6. $0.\dot{5}71428$

Challenging · Set D

1. $0.6, \frac{5}{8}, \frac{2}{3}, 0.67$
2. $0.\dot{1}, 0.\dot{2}, 0.\dot{3}; 0.\dot{7}$
3. 0.142857 repeating: 6 digits
4. halves, quarters, fifths, eighths, tenths

Extension

1. $\frac{7}{9}$
2. $\frac{2}{11}$

Exit ticket

1. Level 1: 0.6
2. Level 2: 0.875
3. Level 3: $0.\dot{3}$

7.10 Rounding decimals (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: 406.3
2. Ex 1b: 125.72
3. Ex 2a: 3.90
4. Ex 2b: 2345.9
5. Ex 3a: 78654.06
6. Ex 3b: 678.431

Easy · Set A (quick drill)

1. 406.3
2. 3.1
3. 12.4
4. 1.0
5. 7.1
6. 2345.9
7. 2.4
8. 4.8
9. 1.0
10. 12.4
11. 7.1
12. 4.0

Easy · Set B

1. 125.72
2. 3.90
3. 0.56
4. 20.00
5. 4.10
6. 78654.06

Medium · Set C

1. 678.431
2. 6
3. 0.05
4. 13.00
5. 0.3
6. 10.0

Challenging · Set D

1. 0.67
2. \$15.74
3. 4.65 and 4.74
4. 3.1, 3.14, 3.142

Extension

1. 23 000
2. 0.0061

Exit ticket

1. Level 1: 5.7
2. Level 2: 12.35
3. Level 3: 0.900

7.11 Decimals and money (booklet p. 43)

WE DO examples (teacher)

1. Ex 1a: \$470.20
2. Ex 1b: \$1.34
3. Ex 2a: \$21.75
4. Ex 2b: \$38.20
5. Ex 3a: \$17.85
6. Ex 3b: \$8.66

Easy · Set A (quick drill)

1. \$8.25
2. \$13.15
3. \$12.00
4. \$6.20
5. \$62.55
6. \$7.80
7. \$4.25
8. \$6.40
9. \$3.60
10. \$2.40
11. \$37.65
12. \$5.10

Easy · Set B

1. \$17.85
2. \$4.10
3. \$10.00
4. \$25.05
5. \$0.50
6. \$13.35

Medium · Set C

1. \$470.20
2. \$1.34
3. \$8.66
4. \$12.00
5. \$42.45
6. \$52.00

Challenging · Set D

1. \$38.20
2. \$44.60
3. 2 L for \$4.60 (\$2.30/L vs \$2.40/L)
4. \$48.40

Extension

1. one of each pack: 8 pens for \$10.70
2. Yes: \$1.20 per pen

Exit ticket

1. Level 1: \$14.60
2. Level 2: \$5.65
3. Level 3: \$12.75